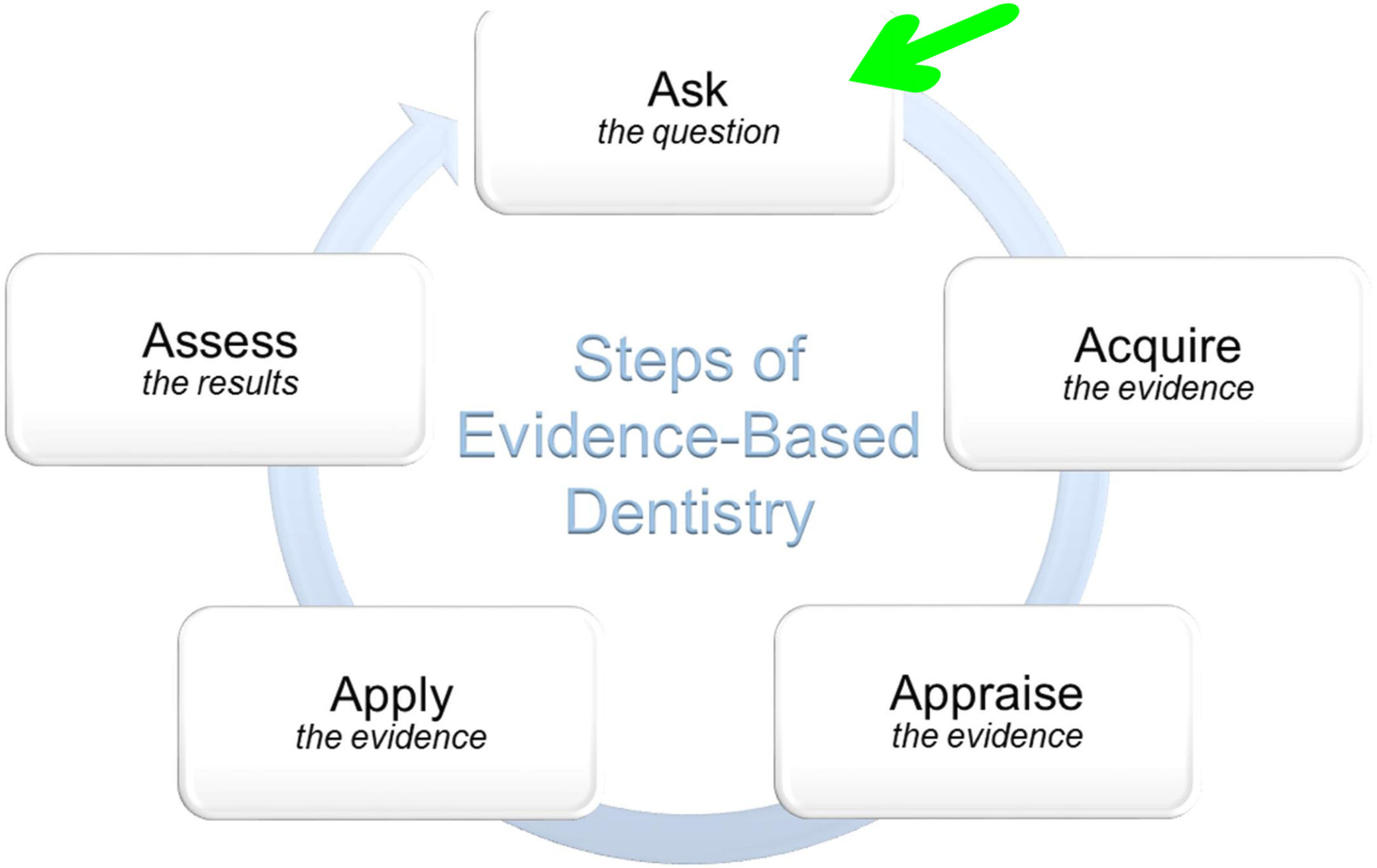
RESEARCH QUESTION

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Ask
the question
Acquire
the evidence
Appraise
the evidence
Apply
the evidence
Assess
the results
Steps of Evidence-Based Dentistry

Steps in EBD



Formulate an answerable question.



Track down the best evidence of outcomes available.



Critically appraise the evidence (ie find out how good it is).



Apply the evidence (integrate the results with clinical expertise and patient values).



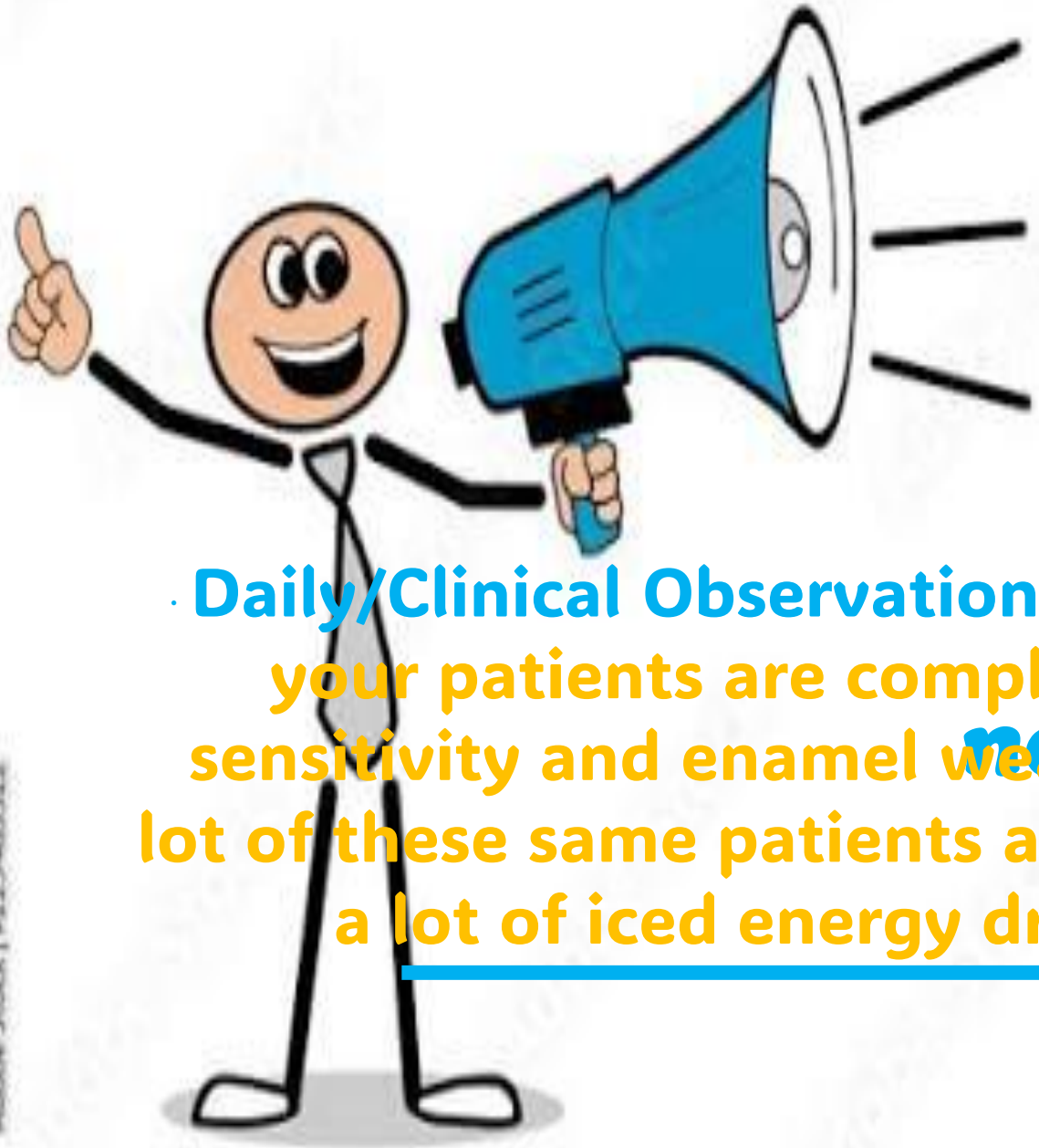
Evaluate the effectiveness and efficiency of the process (to improve next time)

How would you find a research point?

- Through clinical observation
- Through reading to expand your knowledge
- Attend scientific meetings
- Discussions
- Daily observation



The critical thinking



For Example!

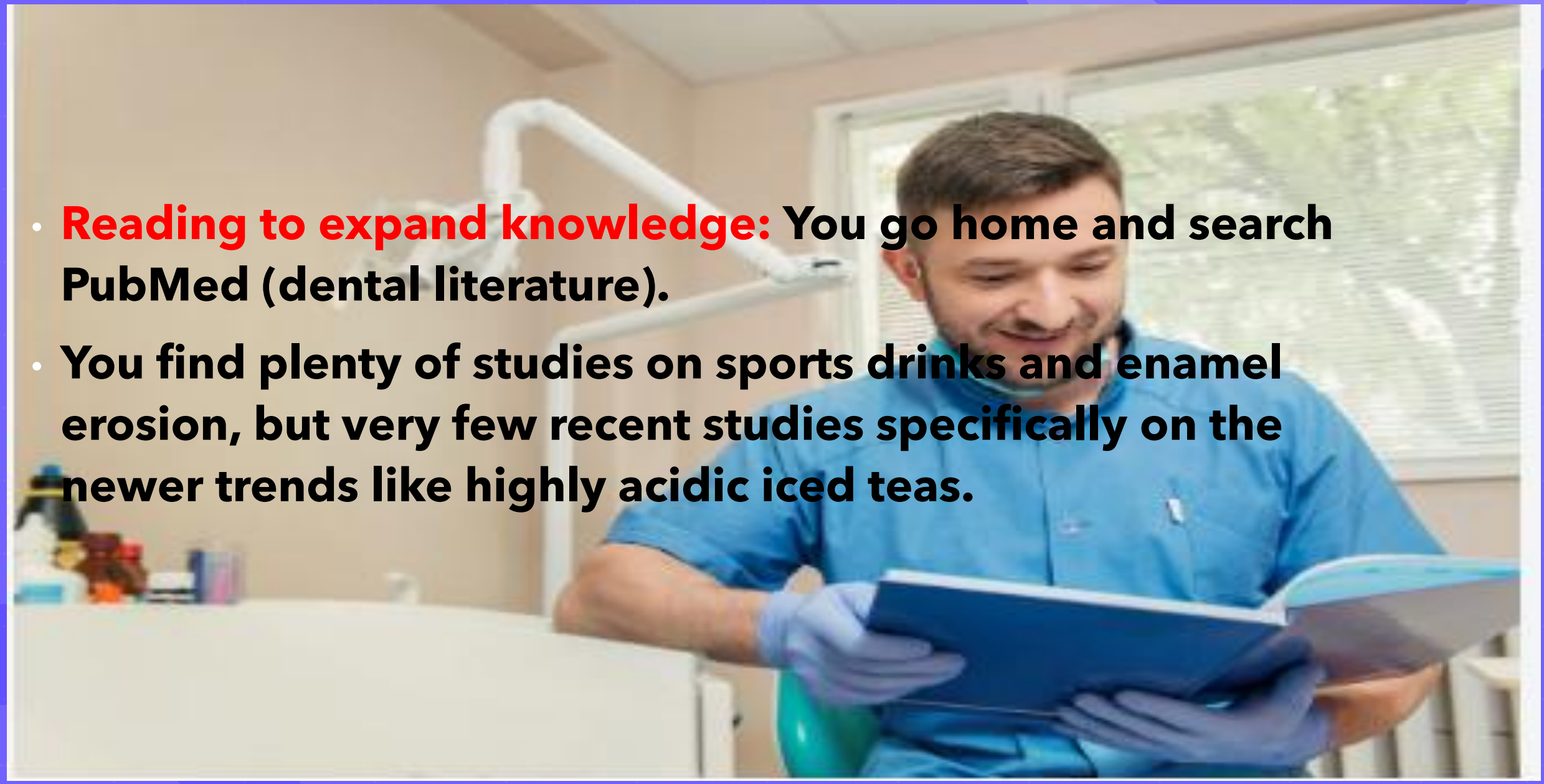
- Daily/Clinical Observation: You notice that many of your patients are complaining of severe tooth sensitivity and enamel wear. You also notice that a lot of these same patients are young adults who drink a lot of iced energy drinks and kombucha.

You are a dentist noticing a frustrati trend in your daily practice.





Discussions: You talk to your dental assistant and a colleague at lunch. Your colleague mentions they've seen the same thing and wonders if it's related to the acidity of these drinks rather than just sugar.



- **Reading to expand knowledge:** You go home and search PubMed (dental literature).
- You find plenty of studies on sports drinks and enamel erosion, but very few recent studies specifically on the newer trends like highly acidic iced teas.



Attend scientific meetings

You go to a dental conference. A speaker gives a lecture on "Modern Dietary Habits and Erosive Tooth Wear." During the Q&A, you ask if anyone has studied kombucha. The speaker says it's an "emerging area of interest."



The Critical Thinking

- You put it all together. You realize there is a gap in the literature. You formulate a research point: "Is there a statistically significant correlation between the daily consumption of kombucha and accelerated enamel erosion in patients aged 18-30 compared to a control group?"

Clinical Research

Framing the question is a key step in the process of searching for evidence to inform clinical decisions

Background Questions

Foundational knowledge, such as definitions and mechanisms of action

Foreground Questions

Therapy and prevention, diagnosis, etiology, and prognosis

Examples of Background questions:

“What is chronic periodontitis?” and “How does the laser caries detection system work?”

Examples of foreground questions:

“Are antibiotics effective as an adjunctive therapy for treating patients with chronic periodontitis?” or “What is the diagnostic accuracy of the laser caries detection system?”

To identify the main components of the question. All types of questions typically have four main components, which vary depending on the nature of the question. The main components of therapy or prevention questions are

- **Population** (the patients relevant to the question);
- **Intervention** (the treatment or prevention strategy, diagnostic test, or exposure that you are interested in (such as a drug, food, surgical procedure, diagnostic test, or exposure to a chemical));
- **Comparison** (the management strategy used as a reference against which to compare the intervention);
- **Outcomes** (the consequences of the intervention we are interested in).

The acronym PICO arises from these four components, and we will refer to these questions as "PICO questions."

P	I	C	O
Population Patient Problem	Intervention Or Exposure	Comparison	Outcome
Who are the patients? What is the problem?	What do we do to them? What are they exposed to?	What do we compare the intervention with?	What happens? What is the outcome?

P	I	C	O	T
Patient/Population/ Problem	Intervention (variable)	Comparison	Outcome	Time











How

P Participants

I Intervention

C Comparison

O Outcome

T Time of follow up

S Study Design

Main components and examples of clinical questions, according to their natures.

NATURE OF THE QUESTION	EXAMPLE	POPULATION	INTERVENTION (EXPOSURE OR DIAGNOSTIC TEST)	COMPARISON (OR REFERENCE STANDARD)	OUTCOMES
Therapy or Prevention	What is the effectiveness of antibiotics in preventing complications such as postoperative infections after third-molar extractions?	Patients undergoing third-molar extractions	Antibiotic prophylaxis	No prophylaxis	Alveolar osteitis, surgical wound infection
Harm or Etiology	Does giving toddlers milk instead of water to drink at night cause caries?	Toddlers	Drinking milk at night	Drinking water at night	Caries
Diagnosis	How useful is a periapical radiograph in detecting interproximal caries?	Patients suspected of having interproximal caries	Periapical radiograph	Bitewing radiograph	Diagnostic accuracy (as assessed by means of true-positive, true-negative, false-positive and false-negative findings)
Prognosis	Are patients with diabetes at higher risk of experiencing complications after third-molar extractions than are patients without diabetes?	Patients undergoing third-molar extractions	Presence of diabetes	Absence of diabetes	Pain, swelling, trismus, postoperative infections

Table 1 Formulating PICO question

Question: example question: What is the effect of antibiotic in preventing pain and complications after root canal therapy in patients with diabetes mellitus?	**Patient/ population**	**Intervention or exposure**	**Comparison**	**Outcome**
	Patients with diabetes mellitus	Use of antibiotics after root canal therapy	No antibiotics/ placebo after root canal therapy	Reduction in pain and complication
Harm or etiology: Does bottle feeding at night cause caries in children?	Children on bottle feeding	Bottle feeding at night	No bottle feeding at night/water consumption only	Incidence of caries
Diagnosis: Is laser fluorescent technique able to diagnose proximal caries more accurately than bitewing radiographs?	Patients with high caries rate	Laser fluorescent	Bitewing radiographs	Diagnosis of proximal caries
Prognosis: Are patients with apical periodontitis at higher risk of failure of root canal therapy than the patients without apical periodontitis?	Patient with and without apical periodontitis	Root canal therapy	NA	Success and failure of the treatment

Abbreviations: NA, not applicable; PICO, Patient/population (P), Intervention (I), Comparison (C), Outcome (O).

Setting an answerable question



How to evaluate a research question?

- **Feasible:** Do you have the expertise, budget and resources to conduct and finish the study on time?
- **Interesting**: Does the study address an important problem in your field? What are the possible future implications of your findings?
- **Novel:** Is the study using a novel method or answering a novel question?
- **Ethical**: Can the question be answered without harming the patient? Do the benefits outweigh the risks?
- **Relevant**: Does the study aim at filling a gap in the literature? Will it guide further research or change clinical management?

PICO Practice Questions



Case 1

The mother of a 5-year-old boy with rampant caries has brought him to the dentist due to severe pain in a previously treated tooth. She wants to know if the restoration can be redone using glass ionomer restoration rather than the present composite one to avoid future recurrence of caries.

Let’s build the PICO

P: ??

Children with rampant caries

I: ??

Glass- ionomer

C: ??

Composite

O: ??

Recurrence of caries

Clinical Question



In a child with rampant caries, does glass-ionomer work as well as composite in preventing recurrent caries?

Case 2

A 25-year-old man with aggressive periodontits comes to you, his dentist, after having extracted 4 teeth (Grade III mobility). He now has 2 lower incisors and 2 upper maxillary molars that exhibit Grade II mobility. He fears that he may have to extract these teeth as well and has come seeking a solution.

Case 2, continued

You make your usual recommendations that he increase oral hygiene measures and perform routine periodontal debridement, but you know the chances of success are not high. You recall hearing about a study that said regeneration using PRF may help in this situation and decide to review the evidence.

Let’s build the PICO

P: ??

Adult males with aggressive periodontitis/ Grade II Mobile teeth in patients with aggressive periodontitis

I: ??

Regeneration using PRF

C: ??

Conventional periodontal debridement

O: ??

Reduction of mobility

Clinical Question



In mobile teeth in patients with aggressive periodontitis, does PRF result in reduced tooth mobility?

Case 3

10-year-old boy who has an appointment with you for treatment of caries. After the completion of the visit the father asked you: “Doctor, we see a lot of ads about sugar-free chewing gums. Are they good for my child”?

Let’s build the PICO

P: ??

Children

I: ??

Sugar-Free: Xylitol or sorbitol

C: ??

Known standard of care: No xylitol, fluoride

O: ??

Good: Lower caries incidence

Clinical Question



In Children, does sugar-free gum / Xylitol gum lower the incidence of caries?

Do children with deciduous teeth who consume xylitol chewing gum have a lower incidence of dental caries compared to children who don't?

Case 4

A company representative came to your clinic to discuss a new powered toothbrush. He claims it is very good for oral hygiene and it is superior to a manual brush.

Let’s build the PICO

P: ??

adult patients

I: ??

powered toothbrush

C: ??

manual toothbrush

O: ??

reduction in plaque and/or gingivitis

Clinical Question



For adult patients, will the use of a powered toothbrush, compared to a manual toothbrush, result in a reduction in plaque and/or gingivitis?

Case 5

A new patient visits you. He has a heart murmur with regurgitation. He has moderate periodontitis and has not been seen by a dentist in many years. In the past, his specific health condition was pre-medicated with antibiotic prophylaxis. However, new evidence reveals that premedication is not necessary. You want to make sure that his periodontal scaling and root planning will not cause bacterial endocarditis.

Let’s build the PICO

P: ??

Cardiac patient with moderate periodontitis

I: ??

No antibiotic prophylaxis

C: ??

Antibiotic prophylaxis

O: ??

Lower the risk of bacterial endocarditis

Clinical Question



For Cardiac patients with moderate periodontitis, will the use of Antibiotic prophylaxis differ from no antibiotic prophylaxis in reducing the risk of bacterial endocarditis?


Thank
You